

Class Test 1 (04 August 2026)

1. [100 marks]

(a) [60 marks] Suppose A is an $m \times n$ matrix and B is an $n \times r$ matrix. Show that

$$\text{rank } (AB) \leq \min \{ \text{rank } (A), \text{rank } (B) \}.$$

(c) [40 marks] Prove that if a matrix of rank k is multiplied in either order (that is, either premultiplied or postmultiplied) by a nonsingular matrix, the rank of the product is k .